

Motivation: The Gap

World Scene

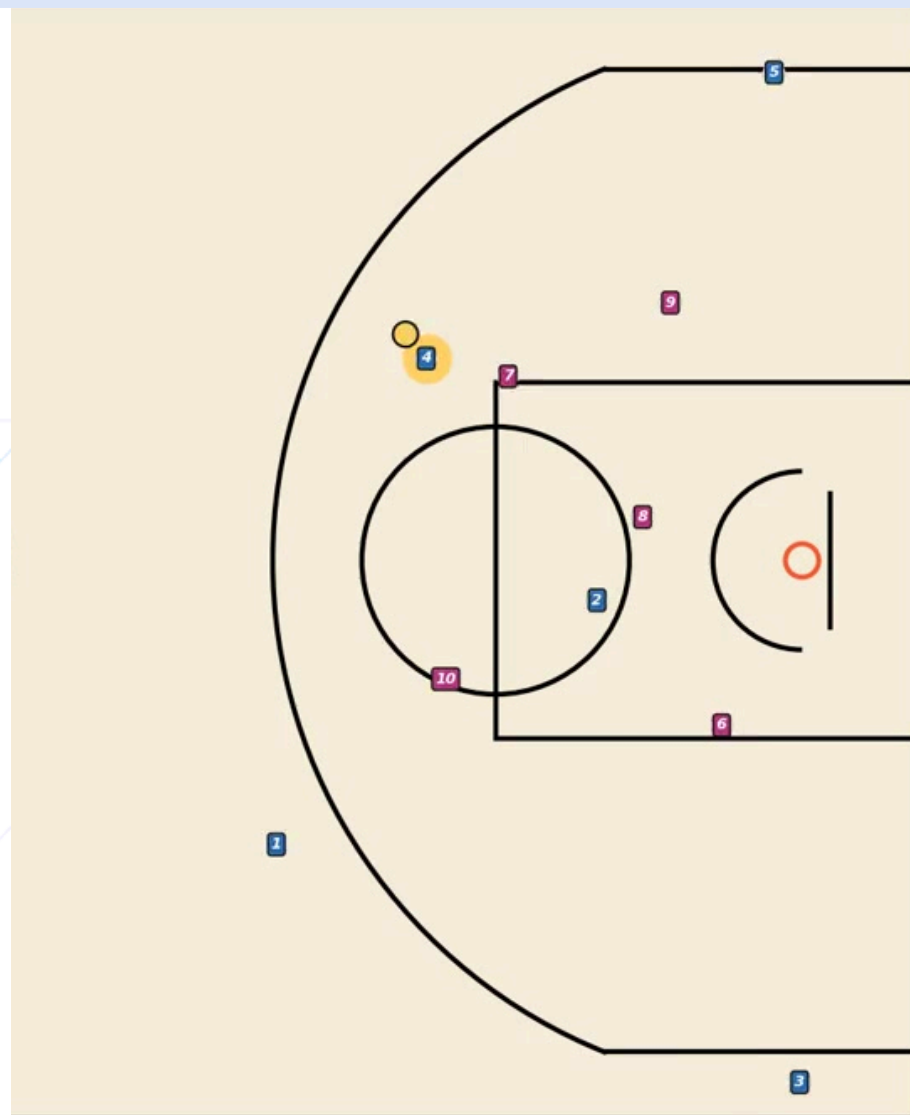
- **Continuous** trajectories: players and ball world (x, y) coordinates
- **Discrete** events: ball possessor
[4 4 0 0 1 1 1 0 0 3 3 3 3 0 0 4 4]

Current Approaches

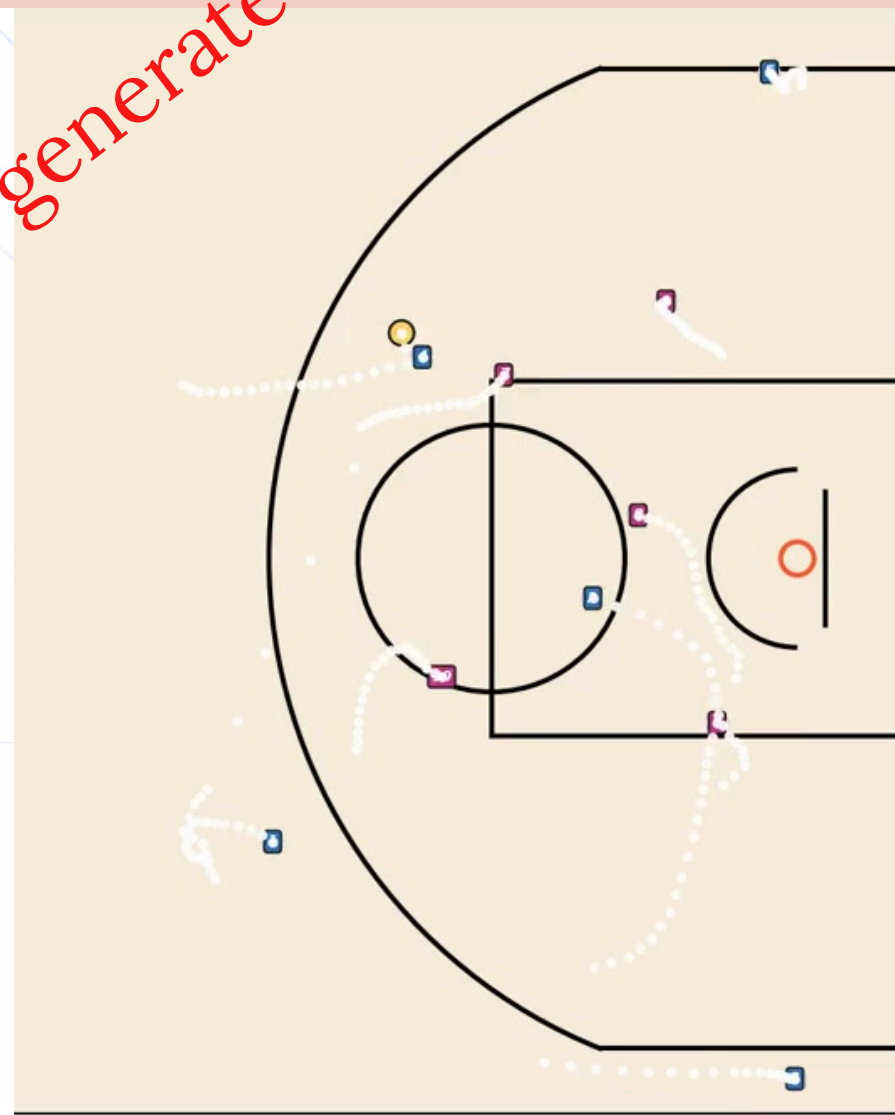
- **Continuous** trajectories modeling, overlook discrete events
- Physically implausible generations (e.g. unrealistic passes)

Our Insight

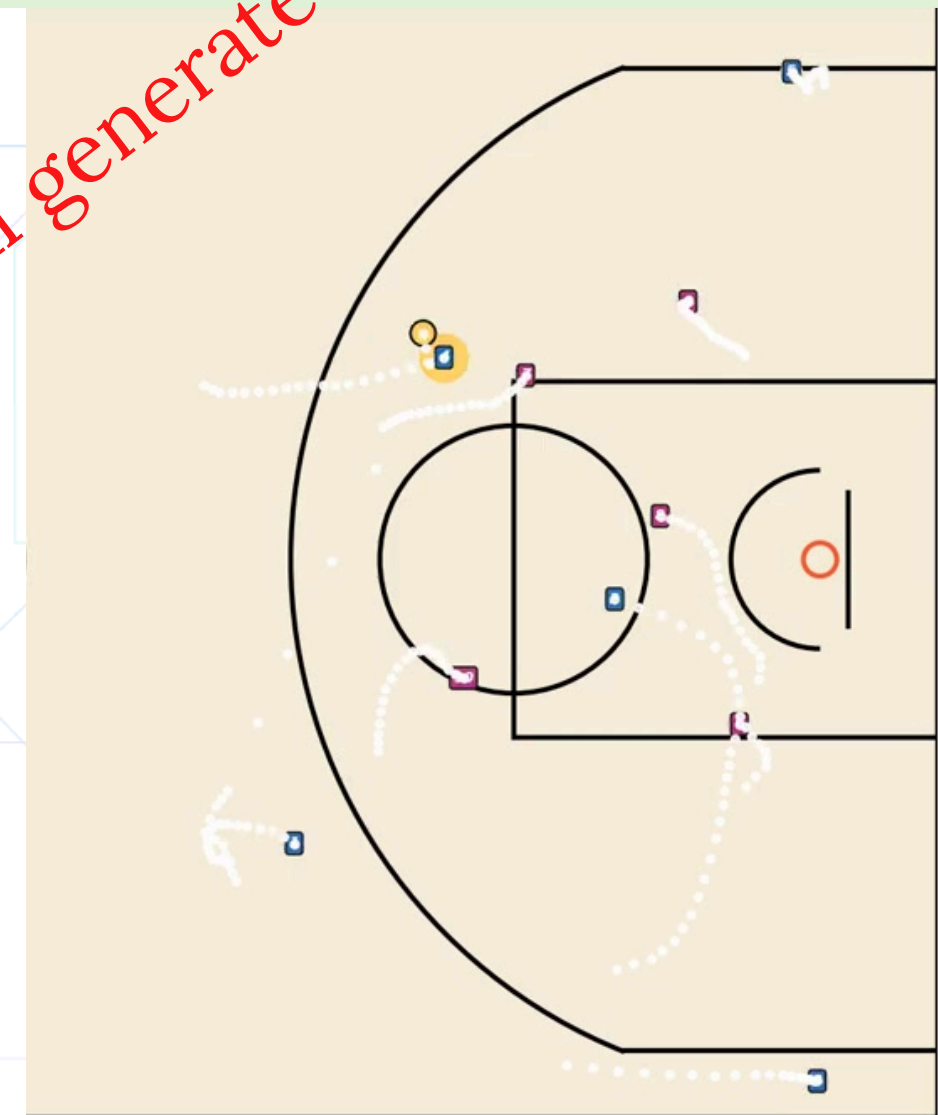
- **Continuous** trajectories & **discrete** events joint modeling
- More **realistic** & **controllable** generation



AI generated



AI generated



● Ball



Home Team



Away Team



Past observations